



COMPUTER NETWORKS

LAB 10



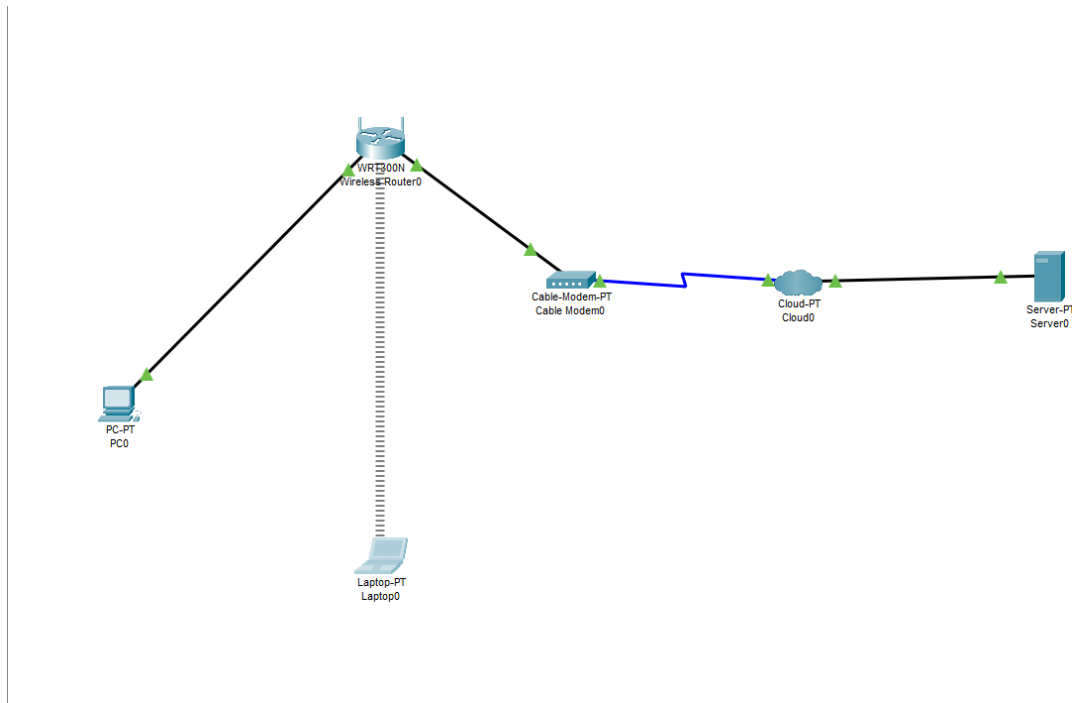
OCTOBER 23, 2025

IFZA RIZWAN
SAP 64281

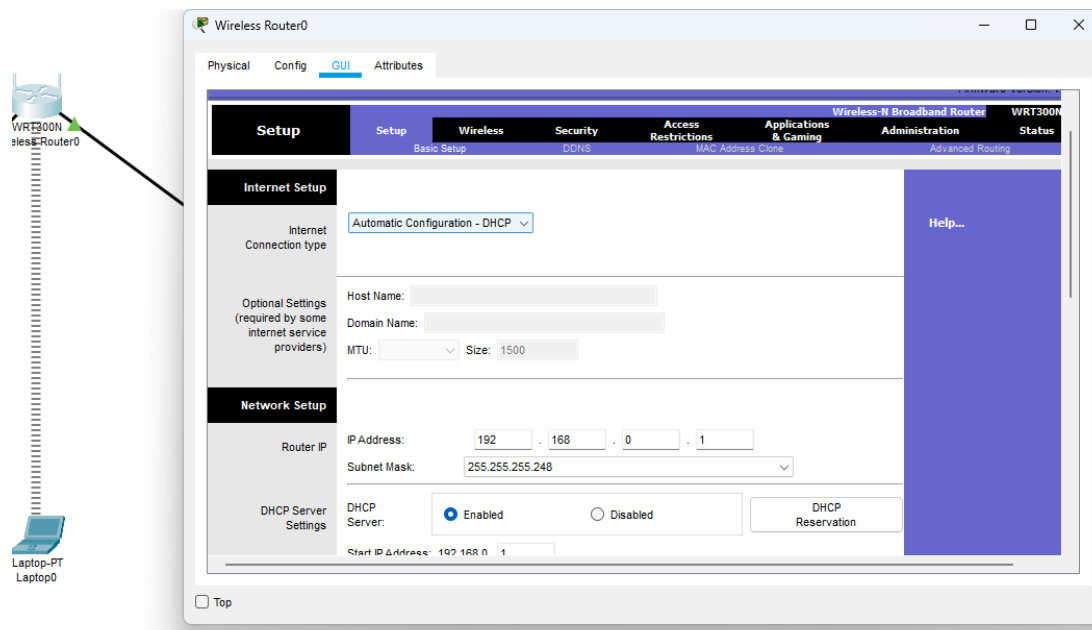
Task 1:

Perform all the tasks in the manual and take screenshot. (10 marks)

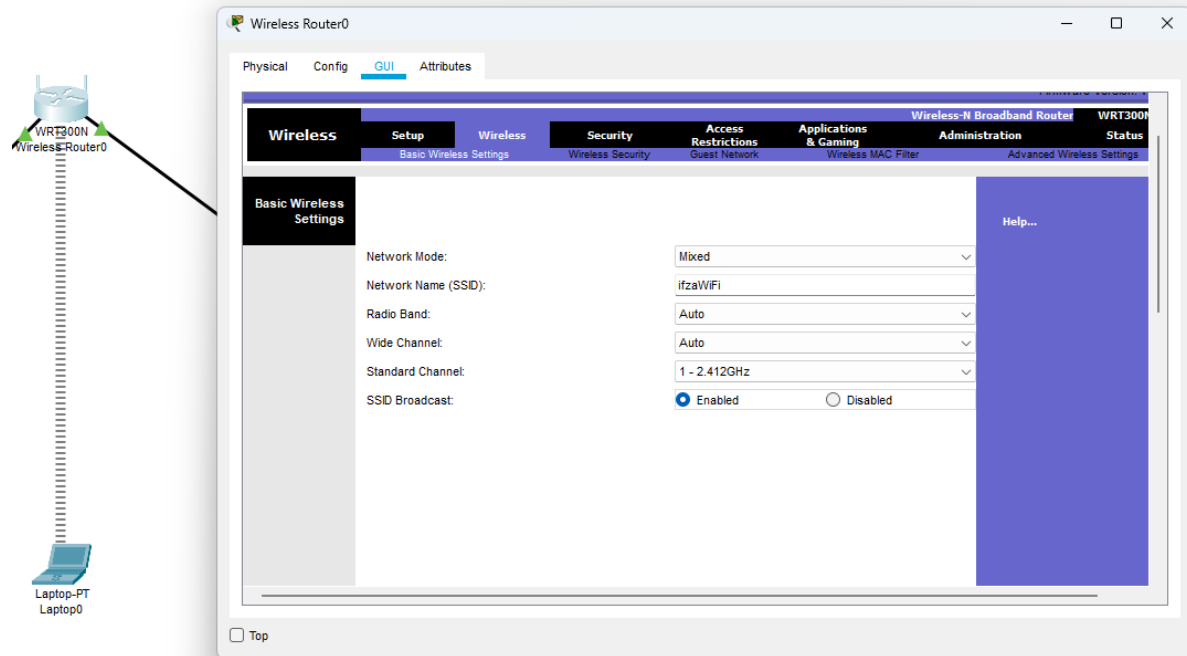
Network:



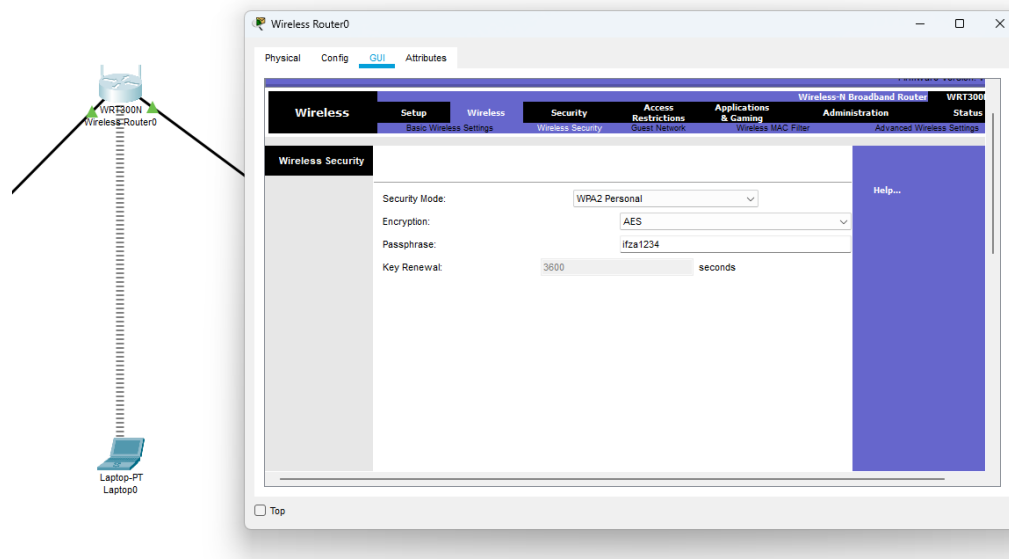
Step one:



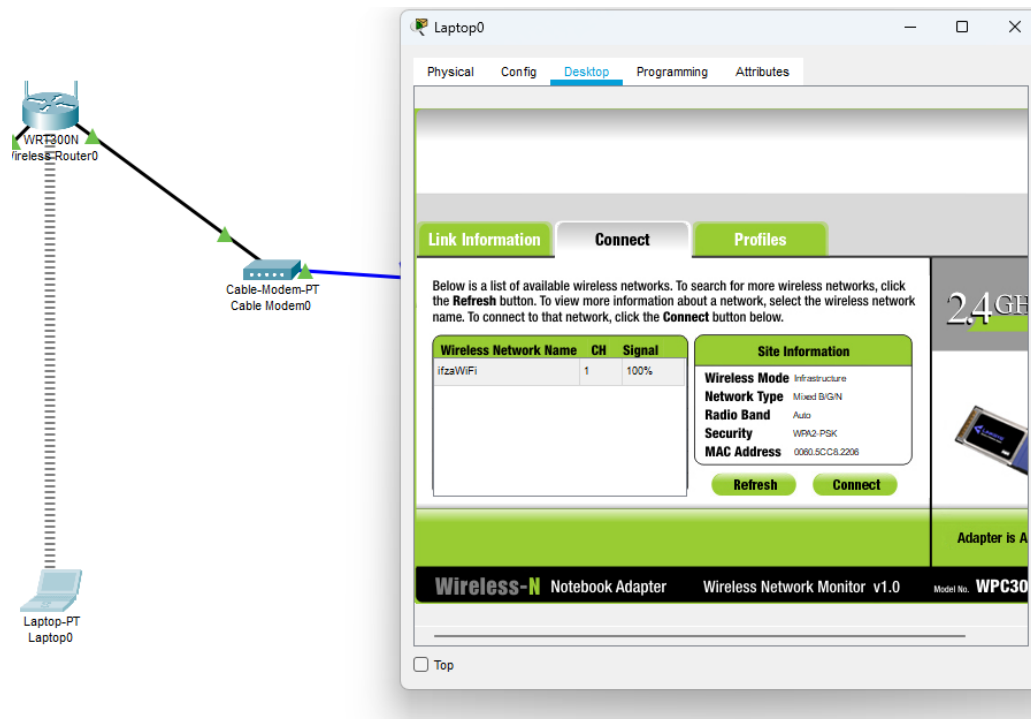
Step 2:



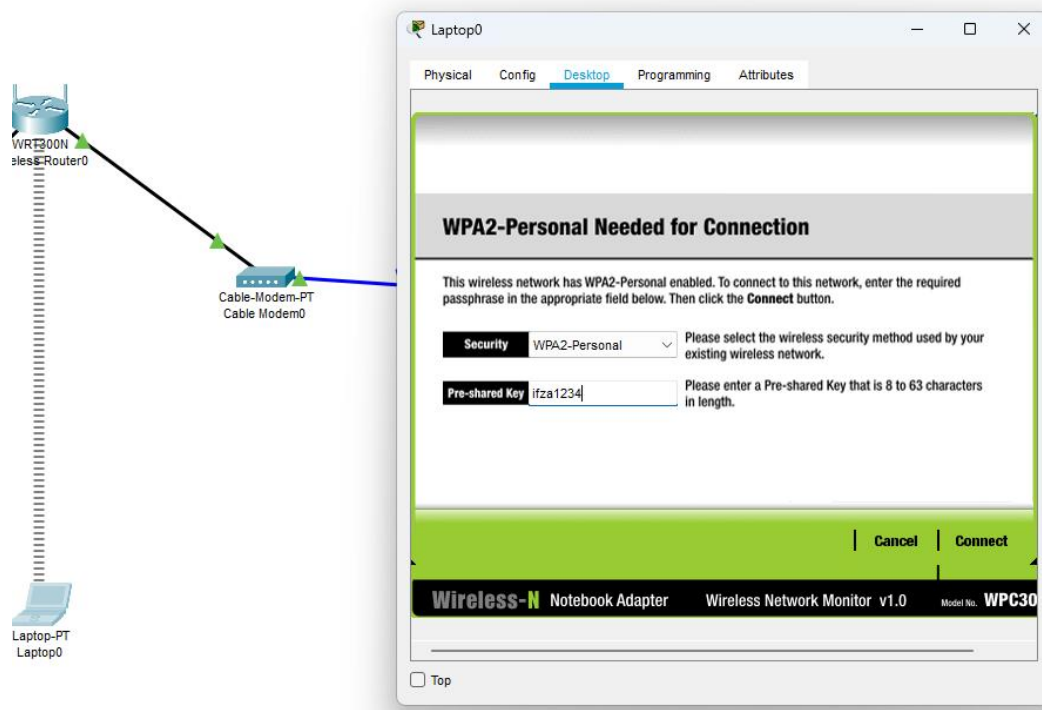
Step 3



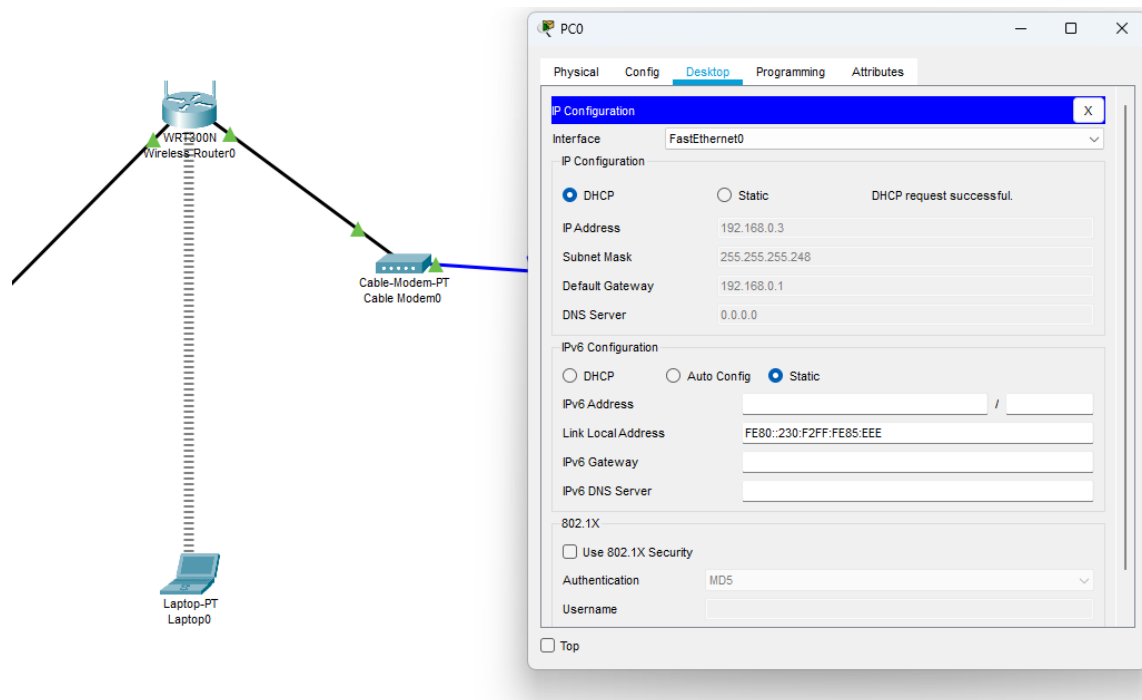
Step 4



step 5



Step 6:

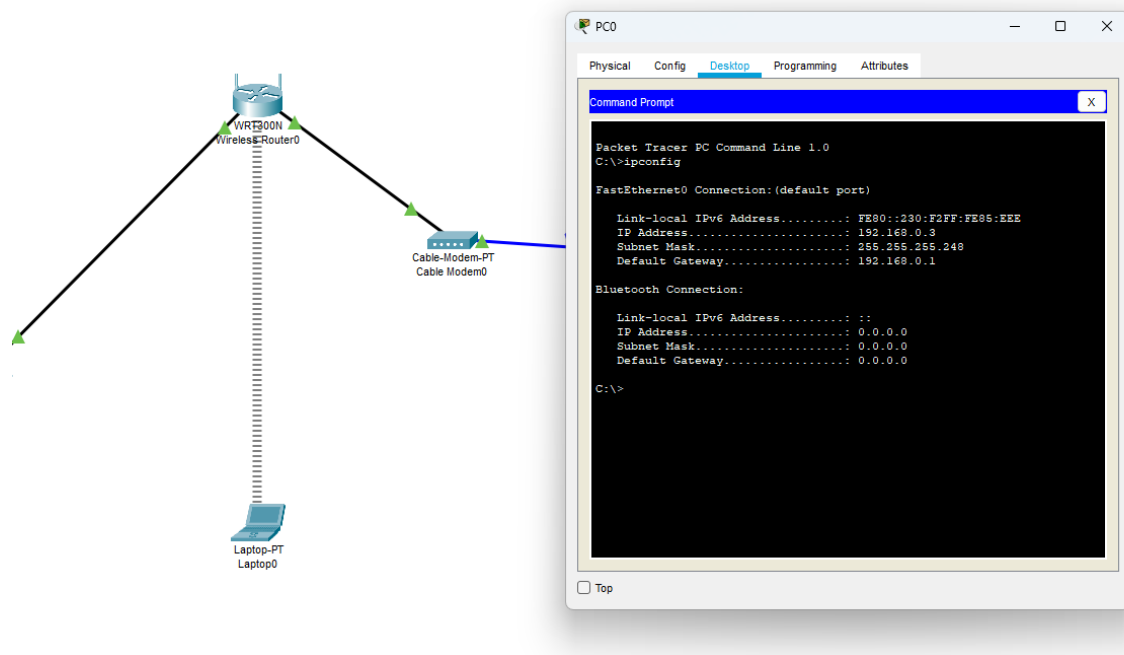


The network diagram shows a Wireless Router0 (WRT300N) connected to a Cable Modem-PT (Cable Modem0) and a Laptop-PT (Laptop0). The PC0 configuration window is open, showing the IP Configuration tab for the FastEthernet0 interface. The DHCP option is selected, and the configuration is successful. The IPv6 Configuration tab is also visible, showing the Static option selected and the Link Local Address set to FE80::230:F2FF:FE85:EEE.

PC0 Configuration Window (Step 6):

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IP Address	192.168.0.3
Subnet Mask	255.255.255.248
Default Gateway	192.168.0.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::230:F2FF:FE85:EEE
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MDS
Username	

Step 7:



The network diagram is the same as in Step 6. The PC0 configuration window is now a Command Prompt window showing the output of the ipconfig command. The output displays the IP configuration for the FastEthernet0 interface, including the Link-local IPv6 Address, IP Address, Subnet Mask, and Default Gateway. The Bluetooth connection information is also shown.

PC0 Command Prompt (Step 7):

```
Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

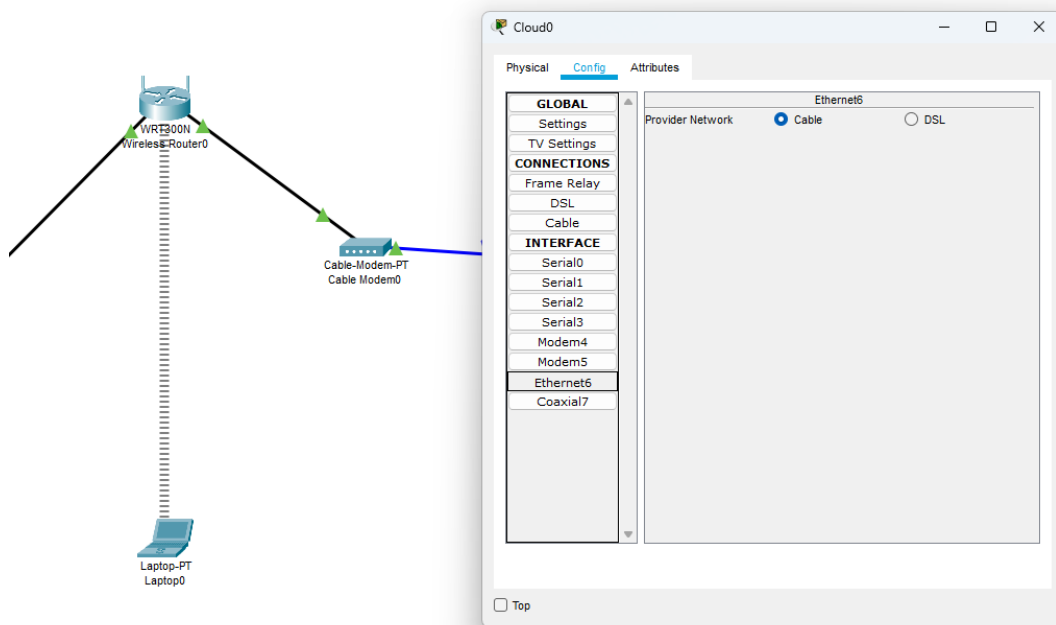
    Link-local IPv6 Address . . . . . : FE80::230:F2FF:FE85:EEE
    IP Address. . . . . : 192.168.0.3
    Subnet Mask . . . . . : 255.255.255.248
    Default Gateway . . . . . : 192.168.0.1

Bluetooth Connection:

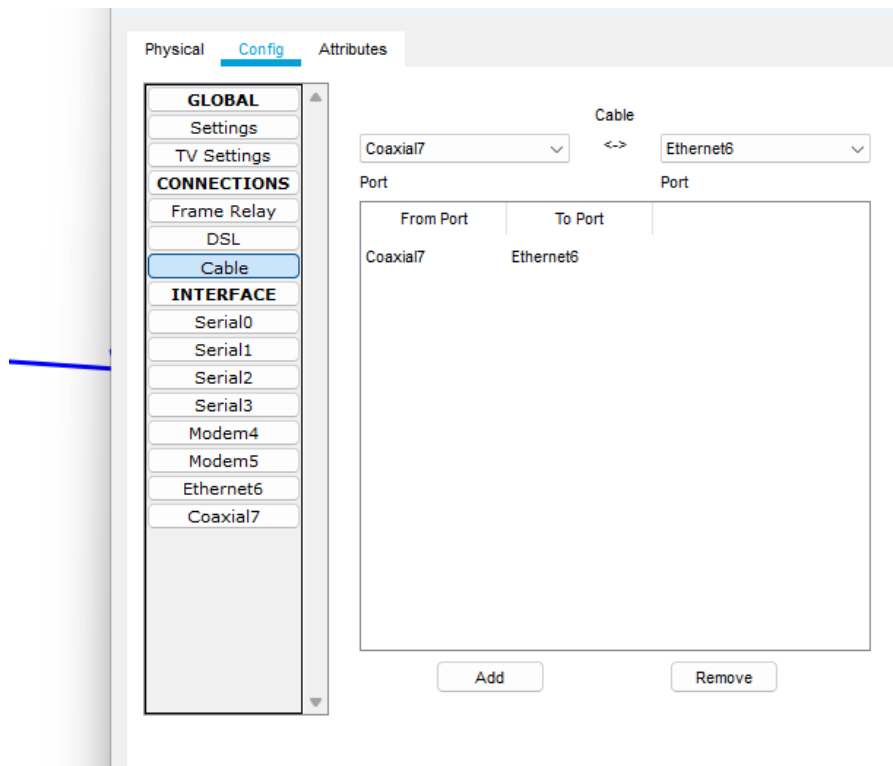
    Link-local IPv6 Address . . . . . : ::
    IP Address. . . . . : 0.0.0.0
    Subnet Mask . . . . . : 0.0.0.0
    Default Gateway . . . . . : 0.0.0.0

C:\>
```

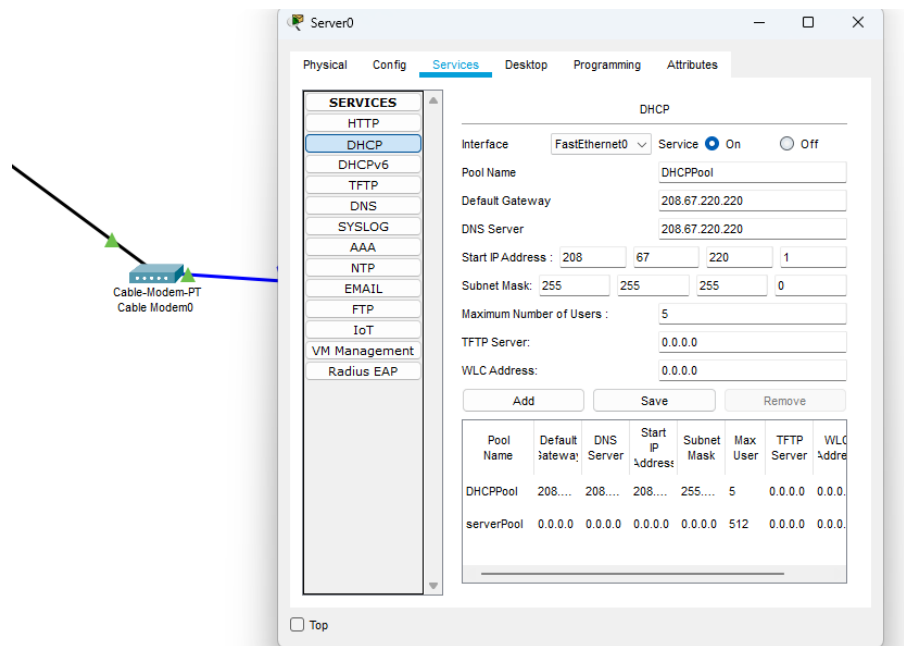
Step 8:



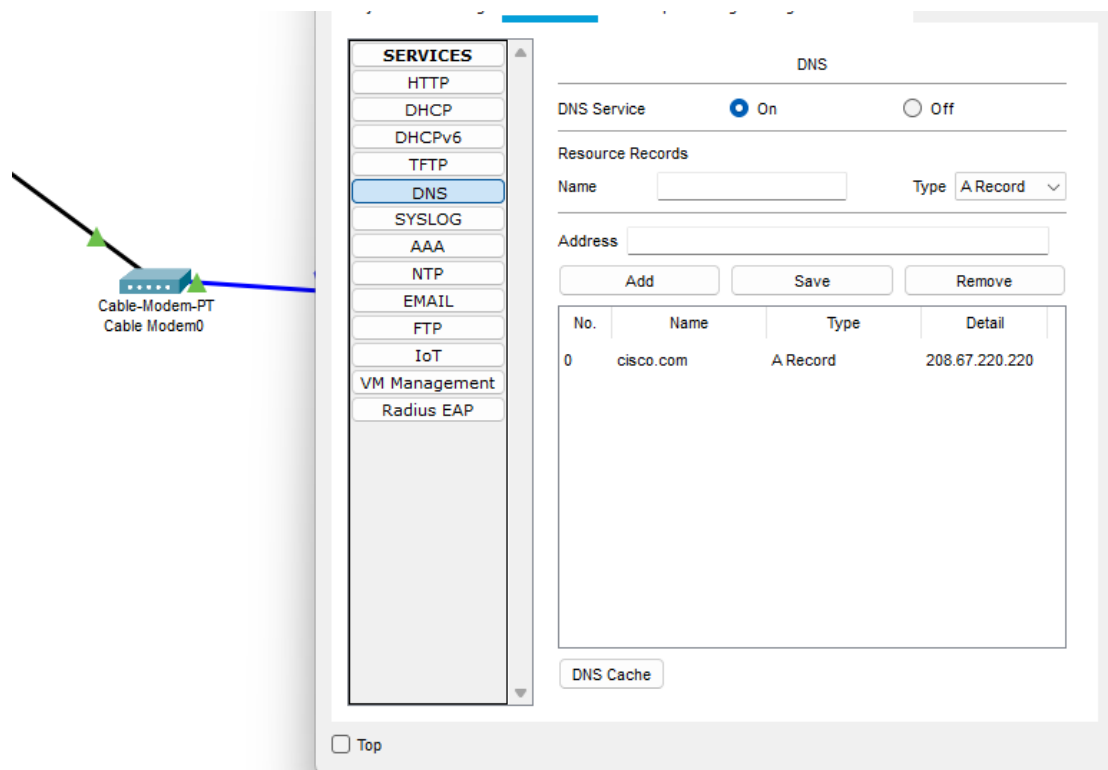
Step 9:



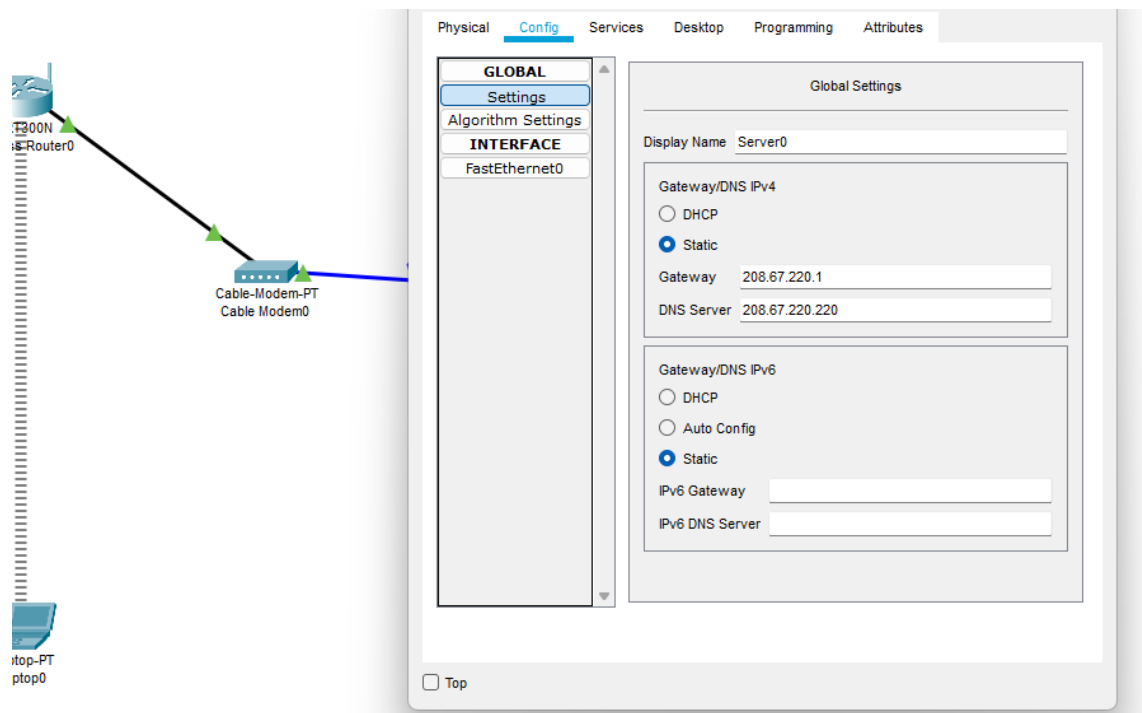
Strep 10:



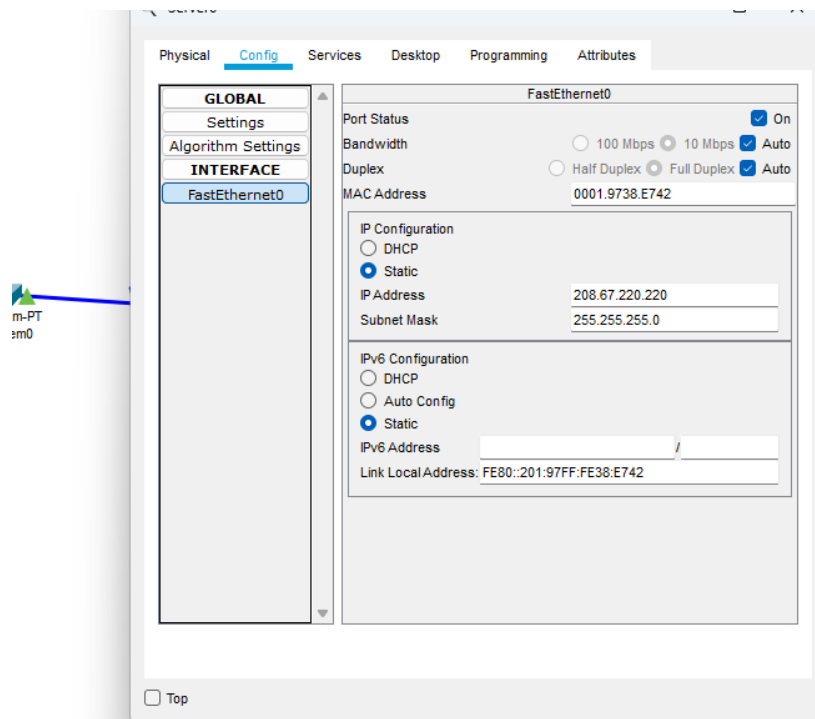
step 11:



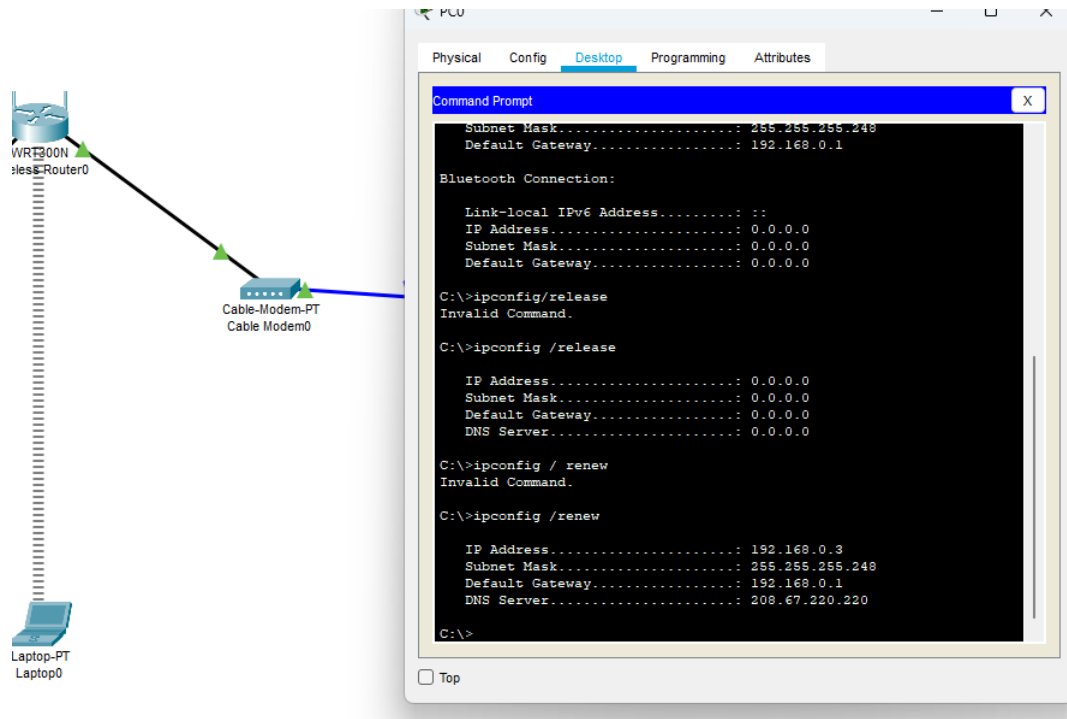
Step 12:



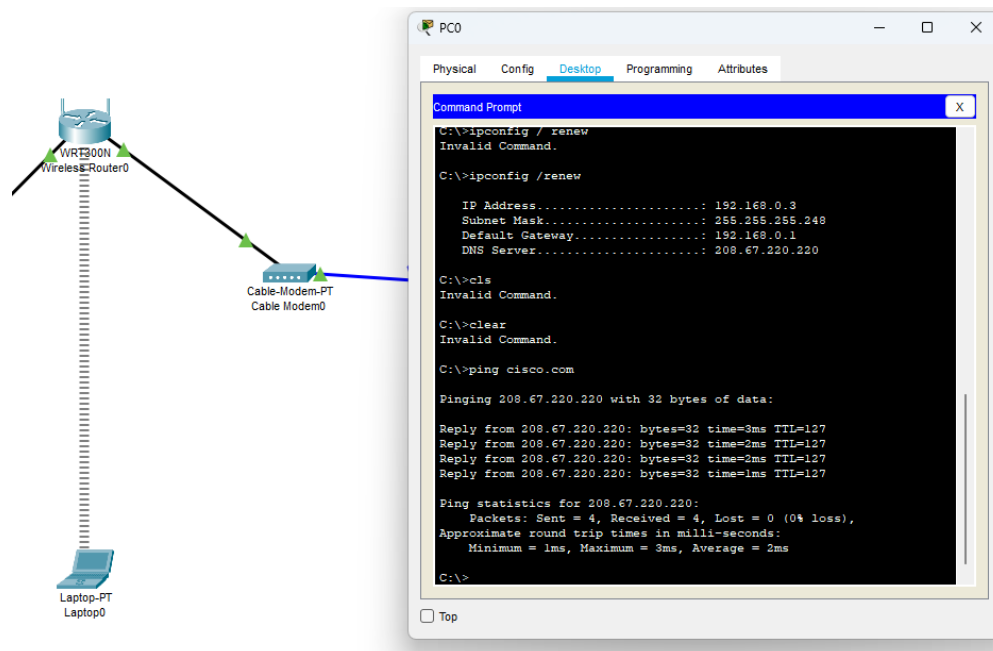
Step 13:

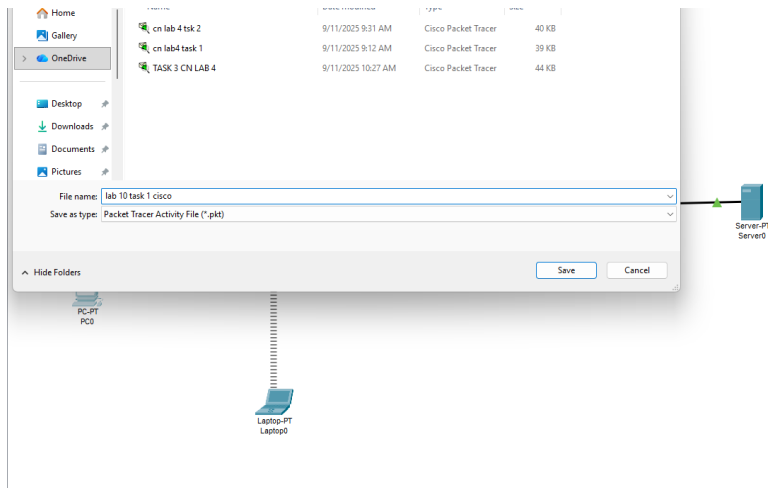


Step 14:

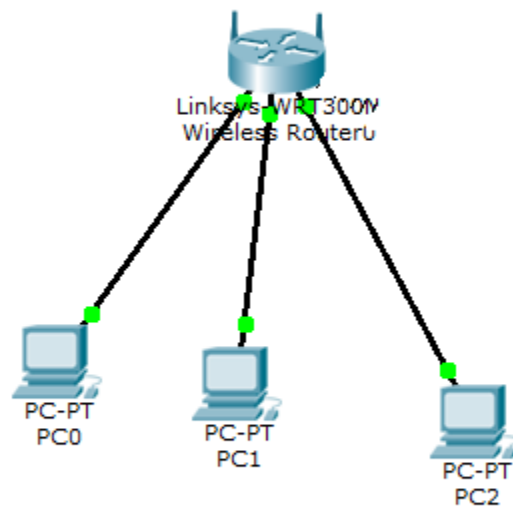


Step 15:





Task 2:



Create the above topology. (10 marks)

Step1:

Firmware Version: v0.93.3

Wireless

Setup

Wireless

Security

Access Restrictions

Applications & Gaming

Wireless-N Broadband Router

WRT300N

Administration

Status

Basic Wireless Settings

Wireless Security

Guest Network

Wireless MAC Filter

Advanced Wireless Settings

Basic Wireless Settings

Help...

Network Mode: Mixed

Network Name (SSID): homewifi

Radio Band: Auto

Wide Channel: Auto

Standard Channel: 1 - 2.412GHz

SSID Broadcast: ☒ Enabled ☐ Disabled

Top

Step 2:

Wireless Router0

Physical

Config

GUI

Attributes

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers): Host Name: Domain Name: MTU: Size: 1500

Network Setup

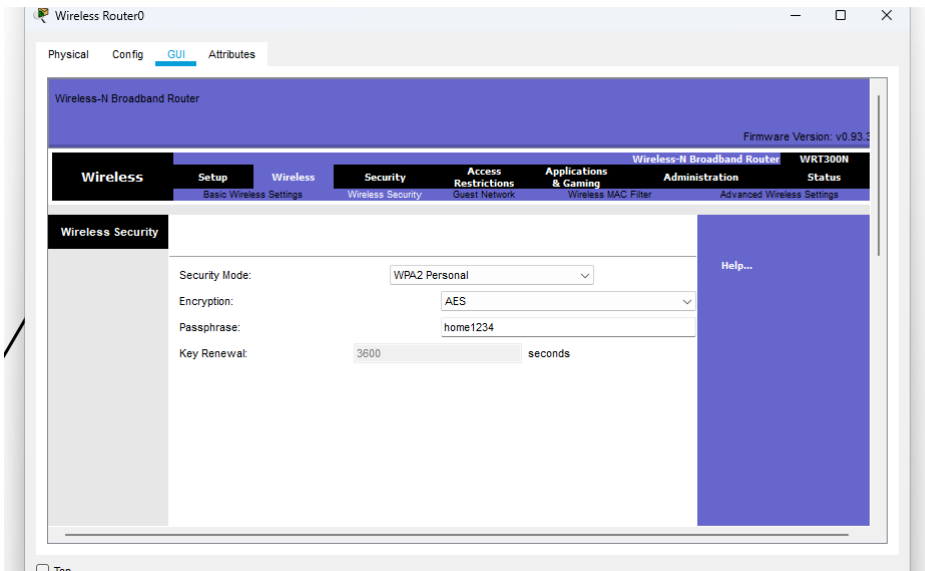
Router IP: IP Address: 192 . 168 . 0 . 1 Subnet Mask: 255.255.255.224

DHCP Server Settings: DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation Start IP Address: 192.168.0. 100 Maximum number of Users: 50 IP Address Range: 192.168.0. 100 - 149

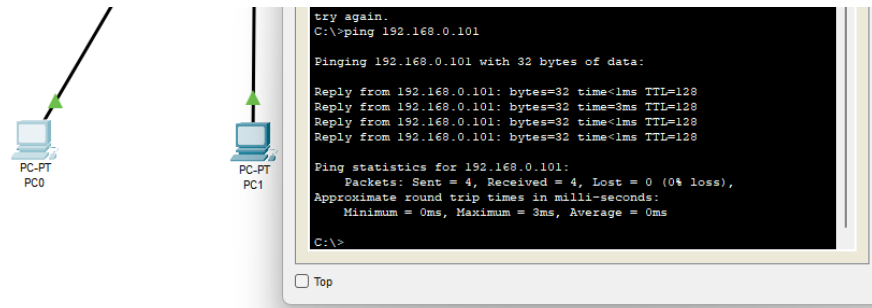
Help...

Top

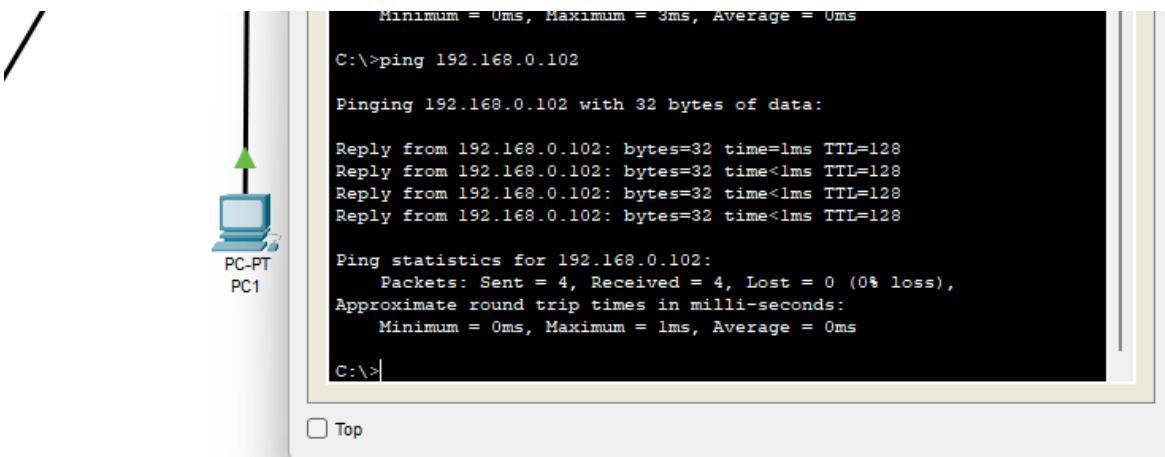
Step 3:



Pc0 to pc1 ping:



Pc0 to pc2 ping:



THE END